Marwadi University
Marwadi Chandarana Group

NAAC A+

TABLETS

Industrial Pharmacy
B. Pharm. Semester 6

Ms. Rashmi Sinha
Assistant Professor
Faculty of Pharmacy, Marwadi University

TABLETS

Introduction

Tablets:

- Tablets are solid dosage forms containing medicaments with or without excipients, prepared by compression or molding methods.
- They are one of the most widely used pharmaceutical dosage forms due to their convenience in administration, accurate dosage, portability, stability, and cost-effectiveness.
- “

Introduction

Tablets:

- "Solid pharmaceutical dosage forms containing drug substances with or without suitable diluents, and prepared either by compression or molding methods."

Ideal Characteristics of Tablets

An ideal tablet should possess the following **characteristics**:

1. Physical Attributes

- **Uniform weight, size, and appearance**
- **Elegant** in appearance and free from cracks, chipping, or discoloration
- Adequate **mechanical strength** to withstand handling

Introduction

2. Chemical and Content Uniformity

- Contain the **correct amount of drug** as stated on the label
- Uniform distribution of active pharmaceutical ingredients (APIs) throughout the batch

3. Biopharmaceutical Properties

- Capable of **disintegrating and dissolving** in the desired time and site of the GI tract
- Maintain **chemical and physical stability** during shelf-life

Introduction

4. Manufacturing Qualities

- Easy to manufacture on a **large scale** using conventional tablet machines
- Acceptable **compressibility** and **flow properties** of granules
- Amenable to coating (if required) without affecting core stability

5. Patient Compliance

- Pleasant appearance, **palatable (if chewable)**, and easy to swallow
- Minimal or no **bitter taste or unpleasant odor** (may be masked)

Classification of Tablets

Oral Tablets for Ingestion

1. Compressed tablets or standard compressed tablets (CT)
2. Multiple compressed tablets (MCT)
3. Layered tablets
4. Compression-coated tablets

Classification of Tablets

5. Chewable tablets Sugar- and chocolate-coated tablets
6. Film-coated tablets
7. Repeat-action tablets
8. Delayed-action and enteric coated tablets
9. Controlled release tablets

Classification of Tablets

Tablets used in the oral cavity

1. Buccal and sublingual tablets
2. Troches and lozenges
3. Dental cones

Tablets administered by other routes

1. Implantation tablets
2. Vaginal tablets

Classification of Tablets

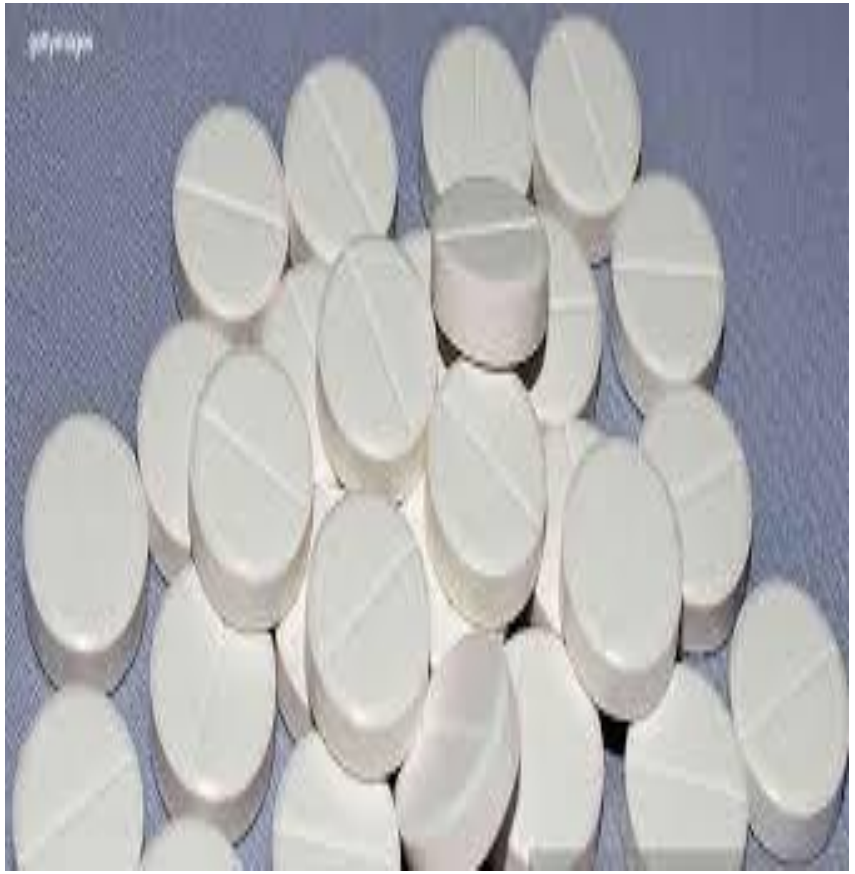
Tablets used to prepare solutions

1. Effervescent tablets
2. Dispensing tablets (DT)
3. Hypodermic tablets (HT)
4. Tablet triturates (TT)

Tablets Ingested Orally

1. Compressed Tablets or Standard Compressed Tablets

This category refers to standard uncoated tablets made by compression and employing any of the three basic methods of manufacture: wet granulation, double compaction, or direct compression



Tablets Ingested Orally

Tablets in this category are :-

- ❖Provide rapid disintegration and drug release.
- ❖Exert a local effect in the gastrointestinal tract
- ❖These drugs are typically water-insoluble and include such therapeutic categories as the antacids and adsorbents.

Tablets Ingested Orally

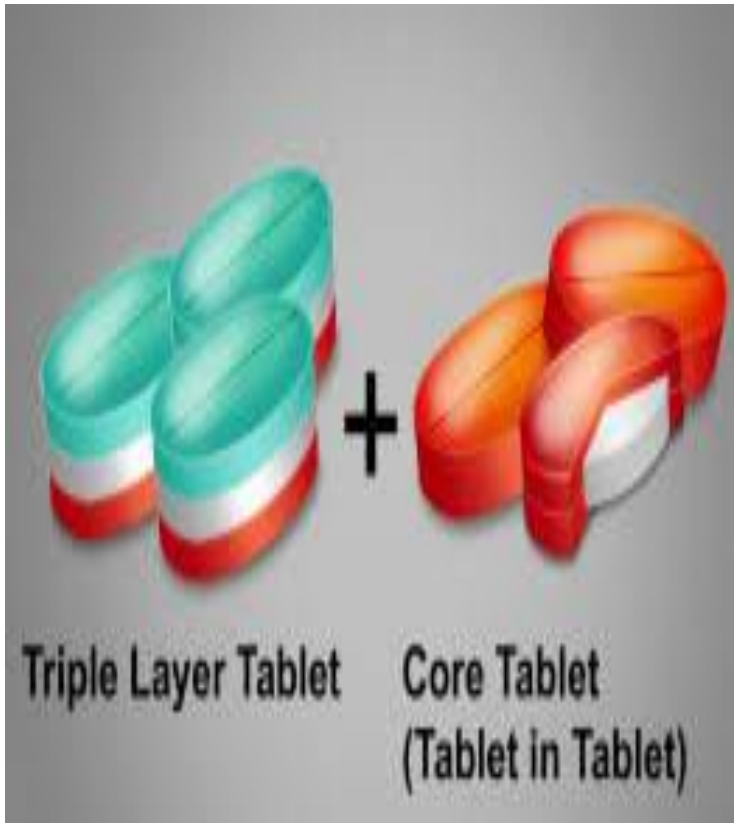
2. Multiple Compressed Tablets

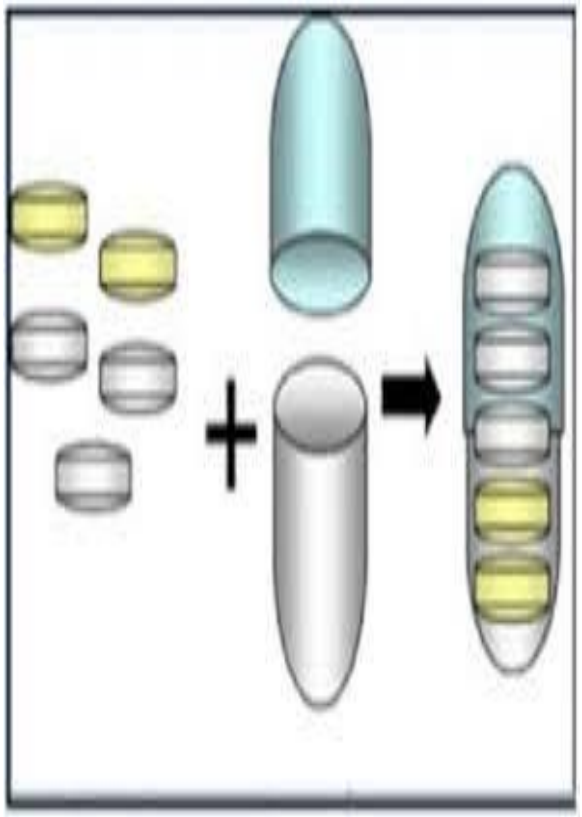
There are two classes of multiple compressed tablets:

Layered tablets and compression-coated tablets.

- Both types may be either two-component or three-component systems: two- or three-layer tablets, a tablet within a tablet, or a tablet within a tablet within a tablet.
- Tablets in this category are usually prepared for one of two reasons: to separate physically or chemically incompatible ingredients, or to produce repeat-action or prolonged-action products.

Tablets Ingested Orally





Tablets Ingested Orally

Multiple compressed tablets :-

- Provide repeat-action wherein one layer of the layered tablet or the outer tablet of the compression-coated tablet provides the initial dose, rapidly disintegrating in the stomach.
- The other layer or the inner tablet is formulated with components that are insoluble in gastric media but are released in the intestinal environment.

Tablets Ingested Orally

3. Chewable Tablets

Chewable tablets are intended to be chewed in the mouth prior to swallowing and are not intended to be swallowed intact.

Tablets Ingested Orally

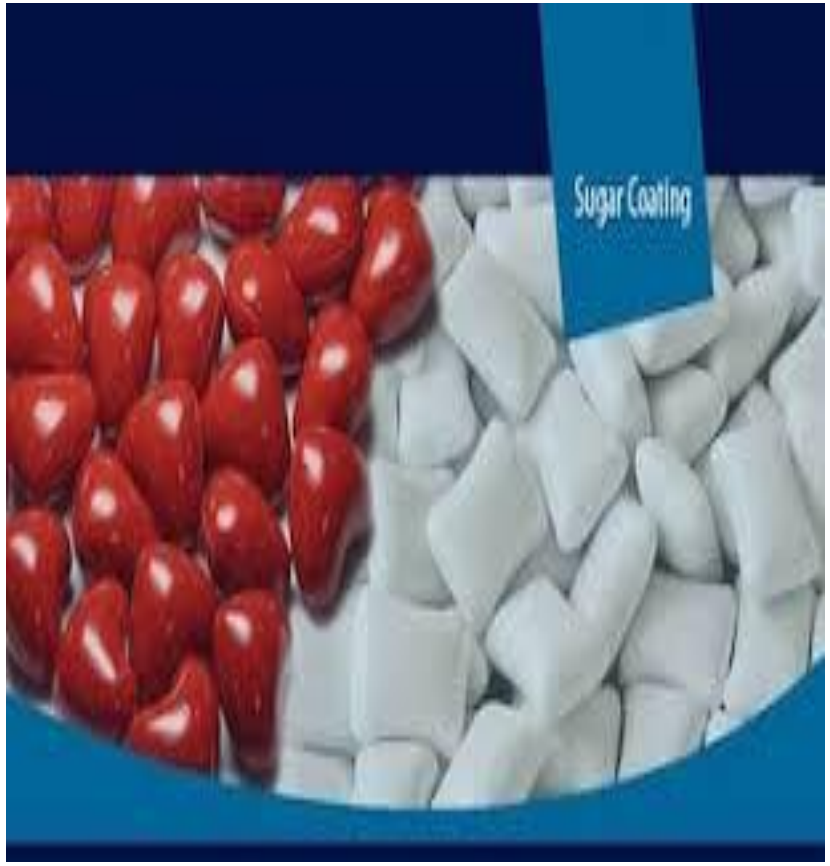
3. Chewable Tablets

The purpose of the chewable tablet is to provide a unit dosage form of medication which can be easily administered to infants and children or to the elderly, who may have difficulty swallowing a tablet intact.

Tablets Ingested Orally

4. Sugar- and Chocolate-coated Tablets

- Their primary historical role was to produce an elegant, glossy, easy-to-swallow tablet dosage form.
- Also, they permit separation of incompatible ingredients between coating and core, and this fact has been widely utilized in preparing many multivitamin and multivitamin mineral combinations



Tablets Ingested Orally

5. Film-coated Tablets:

❖ The initial film-coating compositions employed one or more polymers, which usually included a plasticizer for the polymer and possibly a surfactant to facilitate spreading, all delivered to the tablets in solution from an organic solvent.

Tablets Ingested Orally

- A 30% ethylcellulose dispersion is marketed under the trade name **Aquacoat** by the FMC Corporation.
- Film-coated tablets offer a number of advantages over sugar-coated tablets.
- These advantages include better mechanical strength of the coating based on the elasticity and flexibility of the polymer coating, little increase in tablet weight, the ability to retain debossed markings on a tablet through the thin film coating, the avoidance of sugar.

6. Repeat-action Tablets:

- The mode of operation of repeat-action tablets, and their limitations based on uncontrolled and unpredictable gastric emptying, have just been mentioned.
- In addition to multiple compressed tablets being used for this effect, sugar coated tablets may also be employed.

Classification of Tablets

- The core tablet is usually coated with shellac or an enteric polymer so that it will not release its drug load in the stomach.
- The second dose of drug is then added in the sugar coating, either in solution in the sugar syrup or as a part of the dusting powder added for rapid coat buildup.

Classification of Tablets

7. Delayed-action and Enteric Coated Tablets:

- The delayed-action tablet dosage form is intended to release a drug after some time delay, or after the tablet has passed through one part of the GI tract.
- More recently, polyvinyl acetate phthalate and hydroxypropyl methylcellulose phthalate have come into use. All three polymers have the common feature of containing the dicarboxylic acid, phthalic acid, in partially esterfied form.

Classification of Tablets

- These polymers, being acid esters, are insoluble in gastric media that have a pH of up to about 4; they are intended to hydrate and begin dissolving as the tablets leave the stomach, enter the duodenum (pH of 4 to 6), and move further along the small intestine, where the pH increases to a range of 7 to 8.

Classification of Tablets

- Tablets, are a central component of pharmaceutical dosage forms due to their multiple advantages like stability, convenience, and patient acceptability.
- Understanding their types, manufacturing processes, and ideal properties is crucial for pharmaceutical formulation scientists and industry professionals.
- Tablets can be classified based on:

- **Route of administration**
- **Method of manufacture**
- **Functionality or usage**

Classification of Tablets

1. Based on the Route of Administration

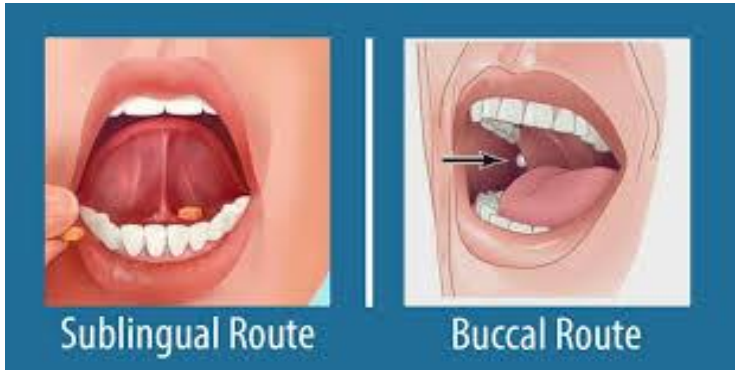
Type	Description
Oral Tablets	Swallowed whole; e.g., uncoated, film-coated, enteric-coated tablets
Buccal Tablets	Placed between cheek and gum for slow absorption
Sublingual Tablets	Placed under the tongue for rapid absorption
Lozenges/Troches	Dissolve slowly in the mouth, for local effect
Effervescent Tablets	Dissolve in water before administration; release CO_2

Classification of Tablets

1. Based on Route of Administration

Oral Tablets

Oral Tablets



Lozenges Tablets

Lozenges Tablets

Effervescent Tablets

Classification of Tablets

3. Based on Coating

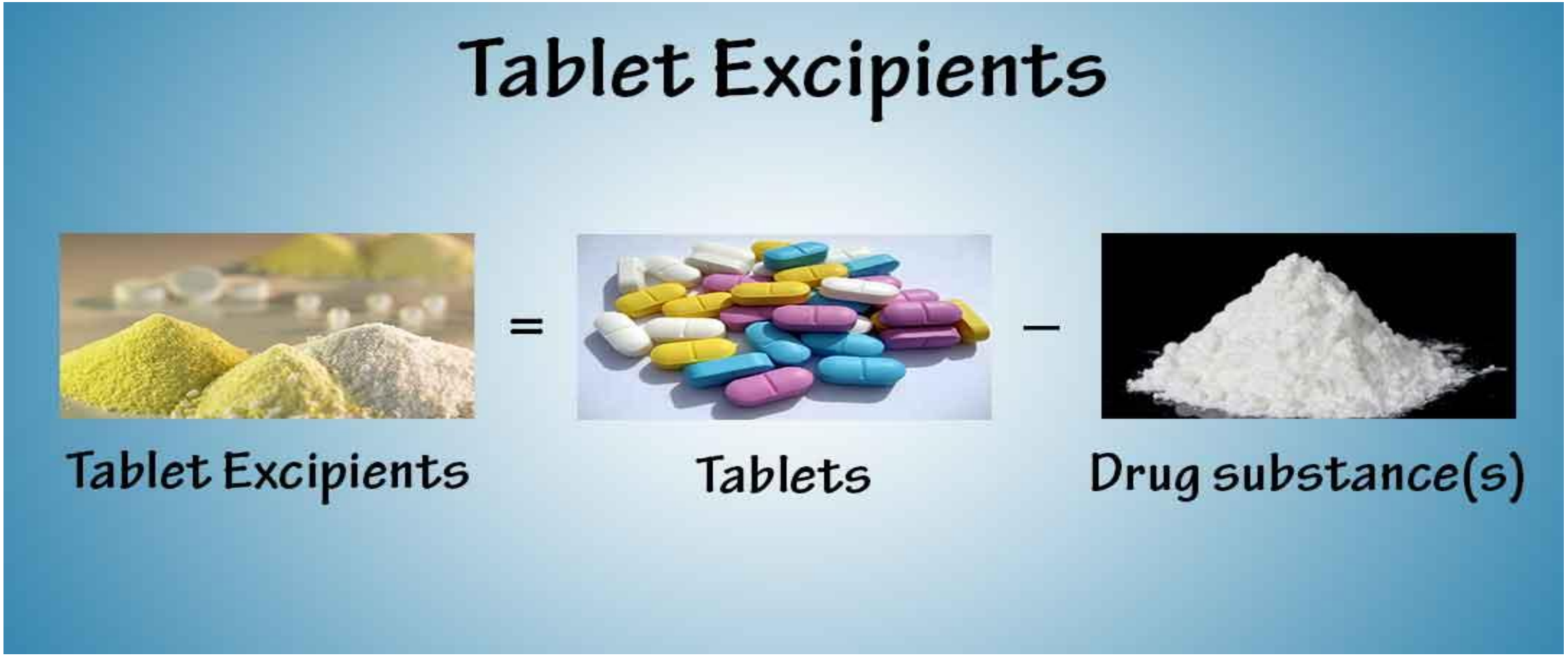
Type	Description
Sugar-coated Tablets (SCT)	Coated with sugar layer for taste masking and appearance
Film-coated Tablets (FCT)	Covered with a thin polymer film
Enteric-coated Tablets (ECT)	Coated to resist stomach acid and dissolve in intestine
Gelatin-coated Tablets	Provide tamper resistance and smoother swallowing

Classification of Tablets

4. Based on Function or Purpose

Type	Description
Chewable Tablets	Designed to be chewed, often for pediatric or geriatric use
Effervescent Tablets	Contain acid and bicarbonate; fizz in water for rapid dissolution
Sustained/Controlled Release Tablets	Release drug over a prolonged period
Multilayer Tablets	Contain two or more layers of different drugs or release profiles
Dispensable Tablets	Meant to be dissolved or dispersed in water prior to administration
Implantable Tablets	Inserted under the skin for long-term drug release

Tablet Excipients



Tablet excipients

Tablet excipients must meet certain criteria in the formulation.

1. They must be nontoxic and acceptable to the regulatory agencies in all countries.
2. They must be commercially available in an acceptable grade.
3. Their cost must be acceptably low.
4. They must not be contraindicated by themselves (e.g. sucrose) or because of a component (e.g. sodium) in any segment of the population.
5. They must be physiologically inert.

Additives Classification

Group I

Impart satisfactory processing & compression characteristics

- Diluents
- Binder
- Disintegrants
- Glidants
- Lubricants

Additives Classification

Group II

- ❑ Gives additional physical property
- ❑ Coloring agent
- ❑ Flavours
- ❑ Sweetening agent

Group III

In case of controlled-release tablets

- ❑ Used solubility retarding materials
- ❑ Polymers
- ❑ Waxes

Formulation of Tablets

Formulating a tablet involves combining the active pharmaceutical ingredient (API) with suitable excipients to produce a solid dosage form that delivers the drug effectively, safely, and consistently.

A. Ingredients Used in Tablet Formulation

1. **Active Pharmaceutical Ingredient (API)** – Therapeutic agent.
2. **Diluents (Fillers):** Provide bulk to the tablet (e.g., lactose, microcrystalline cellulose, starch).

Tablet excipients

3. **Binders:** Provide cohesiveness to the powdered material (e.g., starch paste, PVP, gelatin).
4. **Disintegrants:** Facilitate tablet break-up after administration (e.g., starch, sodium starch glycolate, croscarmellose).
5. **Lubricants:** Reduce friction between wall of tablets and wall of die during tablet ejection (e.g., magnesium stearate). (GPAT, NIPER JEE)

Tablet excipients

6. Antiadherents: prevent adhesion of the tablet material to the surface of die and punches.

7. Glidants: Reduce interparticle friction & improves the flow rate of tablet granulation.

8. Coloring agents: Approved by FDA, colorants are obtained in two form Dye (natural) & Lakes (Synthetic).

Ingredients Used in Tablet Formulation

Diluents:

- Diluents are fillers designed to make up the required bulk of the tablet, when the drug dosage itself is inadequate to produce this bulk.
- The dose of some drugs is sufficiently high that no filler is required (e.g. aspirin and certain antibiotics)
- To provide better tablet properties such as improved cohesion, to permit use of direct compression manufacturing, or to promote flow.

Ingredients Used in Tablet Formulation

Diluents used in Tablet Formulations

Diluent	Proprietary name	Directly compressible
Calcium carbonate*	Cal-Carb, Millicarb, Pharma-Carb, Sturcal	X
Calcium phosphate dibasic*	Cyfos, Calstar, Calipharm, Emcompress	√
Calcium phosphate tribasic*	Tricafos, Tri-Cal, Tri-Tab	√
Calcium sulfate*	Cal-Tab, Compactrol	√

Diluents used in Tablet Formulations

Cellulose, microcrystalline (MCC)	Avicel, Emcocel, Vivacel	√
MCC silicified	Prosolv	X
Dextrates	Emdex	√
Dextrose	Tabfine	X
Fructose	Fructofin	X
Lactitol	Finlac	√
Lactose monohydrate	Fast-Flo, Lactochem, Zeparox Pharmatose, Tablettose	√
Maltitol	Maltisorb, Maltit	X

Ingredients Used in Tablet Formulation

Diluents used in Tablet Formulations

Maltodextrin	Glycidex, Lycatab, Maltrin	√
Maltose	Advantose	X
Mannitol	Pearlitol	√
Sorbitol		√
Starch		X
Starch, pregelatinized	Pharma-Gel, Pre-Jel, Sepistab, Starch 1500, Starx 1500	√
Sucrose		X
Sugar, compressible	Dipac, Nutab	X
Sugar spheres	Nu-Core, Nu-Pareil	X
Talc		X
Xylitol	Xylifin, Xylitab	√

Ingredients Used in Tablet Formulation

Commonly used diluents are: Lactose, Starch, Dextrose, and MCC

Lactose

- Lactose is the most widely used diluent in tablet formulation.
- Lactose is an excipient that has no reaction with most drugs, whether it is used in the hydrous or anhydrous form.

Ingredients Used in Tablet Formulation

- Two grades of lactose are commonly available commercially: a 60 to 80 mesh (coarse) and an 80-to 100-mesh (regular) grade.
- Lactose is a low-cost diluent, but it may discolor in the presence of amine drug bases or salts of alkaline compounds (Milard Reaction)-GPAT, NIPER JEE PYQ

Ingredients Used in Tablet Formulation

Spray-dried lactose:

- Spray-dried lactose is one of several diluents now available for direct compression following mixing with the active ingredient, and possibly, disintegrant and a lubricant.
- It can usually be combined with as much as 20 to 25% of active ingredient without losing these advantageous features.

Ingredients Used in Tablet Formulation

- Spray-dried lactose is especially prone to darkening in the presence of excess moisture, amines, and other compounds, owing to the presence of a furaldehyde.
- A neutral or acid lubricant should be used when spray-dried lactose is employed.

Ingredients Used in Tablet Formulation

Starch

- Starch, which may come from corn, wheat or potatoes, is occasionally used as a tablet diluent.
- The USP grade of starch, however, has poor flow and compression characteristics and possesses a high typical moisture content of between 11 and 14%.
- Various directly compressible starches are now available commercially.

Ingredients Used in Tablet Formulation

- Sta-Rx 1500 is one such free-flowing, directly compressible starch; it may be used as a diluent, binder, and/or disintegrating agent.
- Two hydrolyzed starches are Emdex and Celutab, which are basically 90 to 92% dextrose and about 3 to 5% maltose. They are free-flowing and directly compressible. (GPAT, NIPER JEE PYQ)

Dextrose

- Dextrose is also used as a tablet diluent.
- It is available from one supplier under the name **Cerelose** and comes in two forms: as a hydrate, and in anhydrous form for when low moisture contents are required.

Mannitol

❖Mannitol is perhaps the most expensive sugar used as a tablet diluent, but because of its negative heat of solution, its slow solubility, and its pleasant feeling in the mouth, it is widely used in chewable tablets.

Ingredients Used in Tablet Formulation

- It is relatively nonhygroscopic and can be used in vitamin formulation, in which moisture sensitivity may be a problem.
- Mannitol formulations typically have poor flow characteristics and usually require fairly high lubricant levels.

Sorbitol

Sorbitol is an optical isomer of mannitol and is sometimes combined in mannitol formulations to reduce diluent cost; however, sorbitol is hygroscopic at humidities above 65%.

Ingredients Used in Tablet Formulation

Sucrose, or sugar, and various sucrose-based diluents are employed in tablet making.

Some of the sucrose based diluents have such tradenames as:

1. **Sugartab (90 to 93% sucrose plus 7 to 10% invert sugar)**
2. **DiPac (97% sucrose plus 3% modified dextrins)**
3. **Nu Tab (95% sucrose and 4% invert sugar with a small amount of corn starch and magnesium stearate).** (GPAT, NIPER JEE PYQ)

Microcrystalline cellulose

- Microcrystalline cellulose, often referred to by the **tradename Avicel,** is a direct compression material.
- Two tablet grades exist: pH 101 (powder) and pH 102 (granules).
- The flow properties of the material are generally good, and the direct compression characteristics are excellent.

Binders and Adhesives

❖These materials are added either dry or in liquid form during wet granulation to form granules or to promote cohesive compacts for directly compressed tablets.

Examples: Acacia, tragacanth, Strach, and Gelatin

Acacia and tragacanth

- Acacia and tragacanth are natural gums employed in solutions ranging from 10 to 25% concentration, alone or in combination.
- When these materials are used, their wet granulation masses should be quickly dried at a temperature above 37° to reduce microbial proliferation.

Gelatin

- Gelatin is a natural protein and is sometimes used in combination with acacia.
- It is a more consistent material than the two natural gums, is easier to prepare in solution form, and forms tablets equally as hard as acacia or tragacanth.

Starch

- Starch paste has historically been one of the most common granulating agents.
- It is prepared by dispersing starch into water, which is then heated for some prescribed time.
- During the heating, the starch undergoes hydrolysis to dextrin and to glucose

Ingredients Used in Tablet Formulation

Binder used in Tablet Formulations

Binder	Proprietary name	Concentration used (%)
Acacia mucilage		Up to 20 1–5
Alginic acid		
Carbomer	Carbopo	l5–10
Carboxymethylcellulose sodium	Nymcel	5–15
Cellulose, microcrystalline	Avicel, Emcocel, Vivacel	
Ethyl cellulose	Aquacoat	1–3
Gelatin		5–20
Glucose, liquid		Up to 50
Guar gum		1–10

Ingredients Used in Tablet Formulation

Hydroxyethyl	Cellosize	2–6
Hydroxypropyl cellulose	Klucel	2–6
Hydroxypropylmethyl cellulose	Methocel, Pharmacoat	2–5
Magnesium aluminum silicate	Pharmasorb, Veegum	2–10
Maltodextrin	Glucidex, Lycatab, Maltrin	2–10
Methylcellulose	Celacol, Methocel	1–5
Polydextrose	Litesse	
Polyethylene oxide	Polyox	5
Povidone	Kollidon, Plasdone	0.5–05
Sodium alginate	Manucol	1–3
Starch paste		5–25

Ingredients Used in Tablet Formulation

Disintegrants

Added to a tablet formulation to facilitate its breaking or disintegration when it comes into contact with water in the GIT.

Example:

- Starch- 5-20% of tablet weight.
- Starch derivative – Primogel and Explotab (1-8%)
- Clays- Veegum HV, bentonite 10% level in coloured tablet only
- Cellulose and Cellulose derivatives- Ac- Di-Sol (sodium carboxy methyl cellulose)
- Alginate
- PVP (Polyvinylpyrrolidone), cross-linked

Disintegrants

- A disintegrant is added to most tablet formulations to facilitate a breakup or disintegration of the tablet when it contacts water in the gastrointestinal tract.

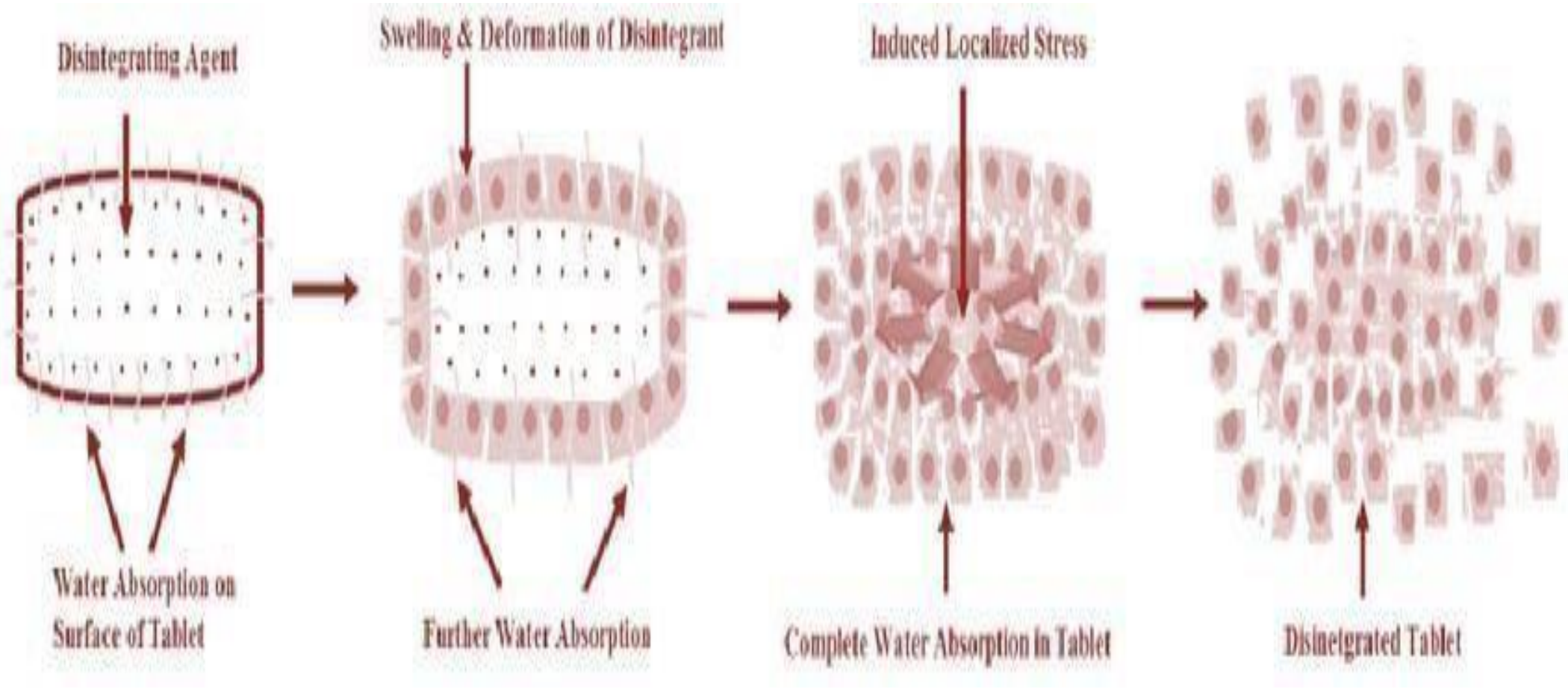
- Disintegrants may function by drawing water into the tablet, swelling, and causing the tablet to burst apart.

Ingredients Used in Tablet Formulation

- The disintegrating agent may be mixed at two stages (i) during the formation of granules, prior to wetting with the granulating fluid-intragranular and (ii) at the second mixing stage during compaction of granules into tablets-extragranular.

- Extragranular disintegrating agents cause the tablet to disintegrate quickly into the granules, while intragranular disintegrants break down the granules, giving a finer product.

Ingredients Used in Tablet Formulation



Ingredients Used in Tablet Formulation

Disintegrants used in Tablet Formulations

Disintegrant	Proprietary name	Concentration used (%)
Alginic acid		2–10
Carboxymethylcellulose sodium		1–5
Cellulose, microcrystalline	Avicel, Emcocel, Vivacel	Up to 10
Croscarmellose sodium*	Ac-di-Sol, Solutab	0.5–5
Crospovidone*	Kollidon CL, Polyplasdone XL	2–5
Docusate sodium		0.5–1
Guar gum		2–8

Ingredients Used in Tablet Formulation

Disintegrants used in Tablet Formulations

Magnesium aluminum silicate	Veegum	2–10
Methylcellulose	Celacol, Methocel	2–10
Polacrilin potassium*	Amberlite	2–10
Poloxamer		5–10
Povidone	Kollidon, Plasdone	0.5–5
Sodium alginate	Manucol	2.5–10
Sodium glycine carbonate		
Sodium lauryl sulfate	Empicol	0.5–2
Sodium starch glycolate*	Explotab, Primojel	2–8
Starch		2–10
Starch, pregelatinized	Lycatab, Pharma-Gel, Pre-Jel, Sepistab,	5–10

* Used as superdisintergrant

Ingredients Used in Tablet Formulation

Disintegrants used in Tablet Formulations

Magnesium aluminum silicate	Veegum	2–10
Methylcellulose	Celacol, Methocel	2–10
Polacrilin potassium*	Amberlite	2–10
Poloxamer		5–10
Povidone	Kollidon, Plasdone	0.5–5
Sodium alginate	Manucol	2.5–10
Sodium glycine carbonate		
Sodium lauryl sulfate	Empicol	0.5–2
Sodium starch glycolate*	Explotab, Primojel	2–8
Starch		2–10
Starch, pregelatinized	Lycatab, Pharma-Gel, Pre-Jel, Sepistab,	5–10

* Used as superdisintergrant

Ingredients Used in Tablet Formulation

Super-Disintegrants:

Swell up to ten fold within 30 seconds when contact with water.

Example:

- Crosscarmellose- cross-linked cellulose,
- Crosspovidone- cross-linked povidone (polymer),
- Sodium starch glycolate- cross-linked starch.
- These cross-linked products swell up to 10 fold within 30 seconds when in contact with water.
- A portion of disintegrant is added before granulation and a portion before compression, which serve as glidants or lubricant.
- Evaluation of carbon dioxide in effervescent tablets is also one way of disintegration.

Lubricant and Glidants

- Lubricants are intended to prevent adhesion of the tablet materials to the surface of dies and
punches,
- Glidants are intended to promote flow of granules or powder material by reducing the friction between the particles.

Example:

- Lubricants: Stearic acid, Magnesium stearate, Talc, PEG (Polyethylene glycols), Surfactants

- Glidants: Corn Starch – 5-10% conc., Talc-5% conc., Silica derivative - Colloidal silicas such as Cab-O- Sil, Syloid, Aerosil in 0.25-3% conc.

Colouring agent

The use of colors and dyes in a tablet has three purposes:

- Masking of off color drugs
- Product Identification
- Production of more elegant product

- All coloring agents must be approved and certified by FDA. Two forms of colors are used in tablet preparation – FD & C and D & C dyes. These dyes are applied as solution in the granulating agent or Lake form of these dyes. Lakes are dyes absorbed on hydrous oxide and employed as dry powder coloring.

Colouring agent

Example:

- FD & C yellow 6 - sunset yellow
- FD & C yellow 5 - Tartrazine
- FD & C green 3 - Fast Green
- FD & C blue 1 - Brilliant Blue
- FD & C blue 2 - Indigo carmine
- D & C red 3 - Erythrosine
- D & C red 22 - Eosin Y

Ingredients Used in Tablet Formulation

Flavouring agents

For chewable tablet - flavor oil are used.

Sweetening agents:

For chewable tablets: Sugar, mannitol.

- Saccharine (artificial): 500 time's sweeter than sucrose

  Disadvantage: Bitter aftertaste and carcinogenic

- Aspartame (artificial)

  Disadvantage: Lack of stability in presence of moisture.

Ingredients Used in Tablet Formulation

Create a detailed mind map showing the classification of excipients used in tablet formulations.

Your map should include major categories such as:

- Diluents (Fillers)
- Binders
- Disintegrants
- Lubricants & Glidants
- Coating agents
- Colouring & Flavouring agents
- Sweetening agents

Also include **examples and functions** for each category.

Methods of Preparation of Tablets

Granulation

Need:

To prevent segregation of the constituents of the powder mix.

To improve the flow properties of the mix

To improve the compaction characteristics of the mix Other reasons:

Toxic, Slightly hygroscopic, denser.

Methods:

Dry granulation Wet granulation

Effect of granulation method on granule structure.

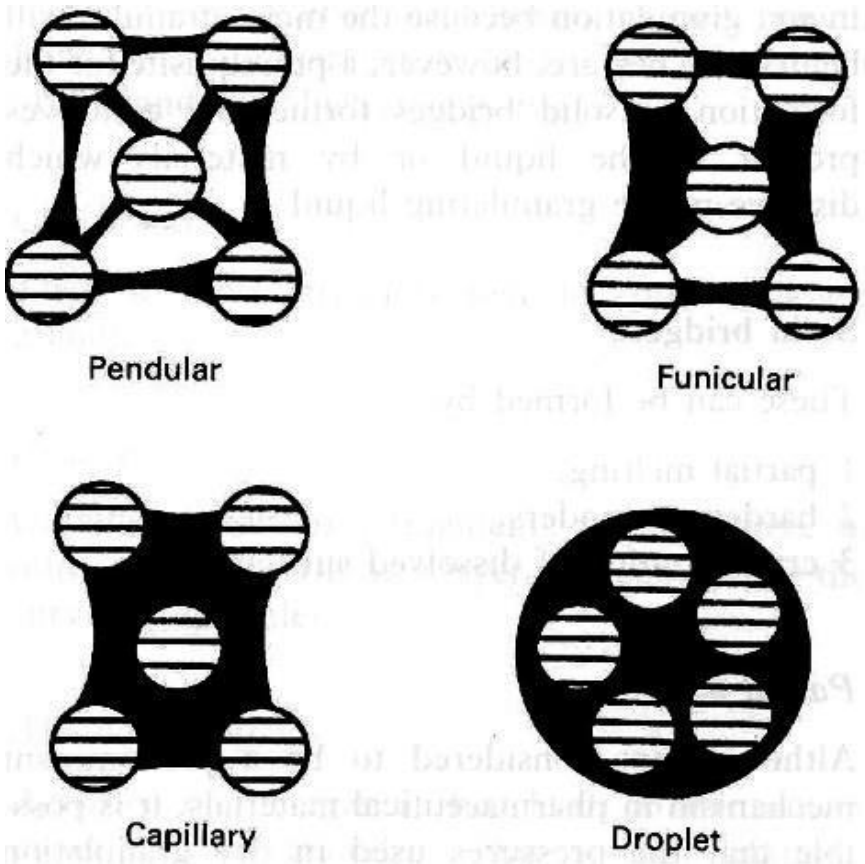
Mechanism of Granulation

There are five particle bonding mechanisms,

1. Adhesion and cohesion forces in the immobile liquid films,
2. Interfacial forces in mobile liquid films within the granules,
3. Formation of solid bridges after solvent evaporation,
4. Attractive forces between solid particles,
5. Mechanical interlocking.

Mechanism of Granulation

- Adhesion and cohesion forces in immobile liquid films between individual primary powder particles.
- Interfacial forces in mobile liquid films.
- Solid bridges: Partial melting, Binder hardening, crystallization of dissolved sub.
- Attractive forces between solid particles.
- Mechanical interlocking.



Granulation Methods

Granulation improves powder flow, compressibility, and content uniformity.

1. Wet Granulation

Definition

Wet granulation involves the **addition of a liquid binder to convert powder blend into granules**.

Principle

Powder particles are **agglomerated using a binder solution**, forming granules that improve flow and compressibility.

Steps involved in Wet Granulation Methods

Steps Involved

1. Weighing & Mixing

- Drug + excipients mixed uniformly

2. Preparation of Binder Solution

- Common binders: starch paste, PVP, gelatin

3. Wet Massing

- Binder solution added to powder blend
- Forms damp mass

Steps involved in Wet Granulation Methods

4. Screening (Wet Sieving)

- Wet mass passed through sieve to form granules

5. Drying

- Tray dryer / Fluidized bed dryer used
- Moisture removed to desired level

6. Dry Screening

- Dried granules passed through sieve

7. Lubrication

- Lubricants added (e.g., magnesium stearate)

8. Compression

- Final tablets prepared by compression

Advantages & Disadvantages of Wet Granulation Methods

Advantages

- Improves **flow properties**
- Enhances **compressibility**
- Produces **uniform and strong tablets**
- Reduces dust formation

Disadvantages

- Not suitable for **moisture-sensitive drugs**
- Not suitable for **thermolabile drugs**
- Time-consuming
- Requires more equipment

Applications of Wet Granulation Methods

Applications

- Most conventional tablets
- Antibiotic tablets
- Vitamins

2. Dry Granulation Method

Definition

Dry granulation is a method in which **granules are prepared without the use of any liquid binder or heat**.

Principle

Powder particles are **compressed to form compacts (slugs or sheets)** which are then broken down into granules.

Steps involved in Dry Granulation Methods

Steps Involved

1. Weighing & Mixing

- Drug + excipients (diluents, disintegrants, lubricants)
- Mixed uniformly

2. Compression (Slugging or Roller Compaction)

Two techniques:

(a) Slugging

- Powder is compressed into large tablets called **slugs** using heavy-duty tablet press.

Steps involved in Dry Granulation Methods

(b) Roller Compaction

- Powder is passed between **two counter-rotating rollers**
- Forms a sheet or ribbon

3. Milling

- Slugs/ribbons are broken into granules using oscillating granulator

4. Sieving

- Granules passed through sieve to obtain uniform size

Steps involved in Dry Granulation Methods

5. Lubrication

. Lubricants and glidants are added

6. Compression into Tablets

. Final compression into tablets using tablet press

Advantages & Disadvantages of Dry Granulation Methods

Advantages

- Suitable for **moisture-sensitive drugs**
- Suitable for **heat-sensitive drugs**
- Fewer processing steps than wet granulation
- No drying step required

Disadvantages

- Requires **specialized heavy equipment**
- Poor compressibility for some drugs
- Risk of **capping and lamination**

Application of Dry Granulation Methods

Applications

- Aspirin tablets
- Effervescent tablets
- Hygroscopic drugs

3. Direct Compression

Definition

Direct compression is a method where **powder blend is directly compressed into tablets without granulation**.

Principle

Uses **directly compressible excipients** that possess good flow and compressibility.

Steps involved in Direct Compression Methods

Steps Involved

1. Weighing

- Drug and excipients weighed accurately

2. Blending

- Mixed uniformly in blender

3. Lubrication

- Lubricants and glidants added

4. Compression

- Directly compressed into tablets

Advantages & Disadvantages of Direct Compression Methods

Advantages

- **Simple and economical**
- Fewer processing steps
- Suitable for **heat and moisture-sensitive drugs**
- Faster production

Disadvantages

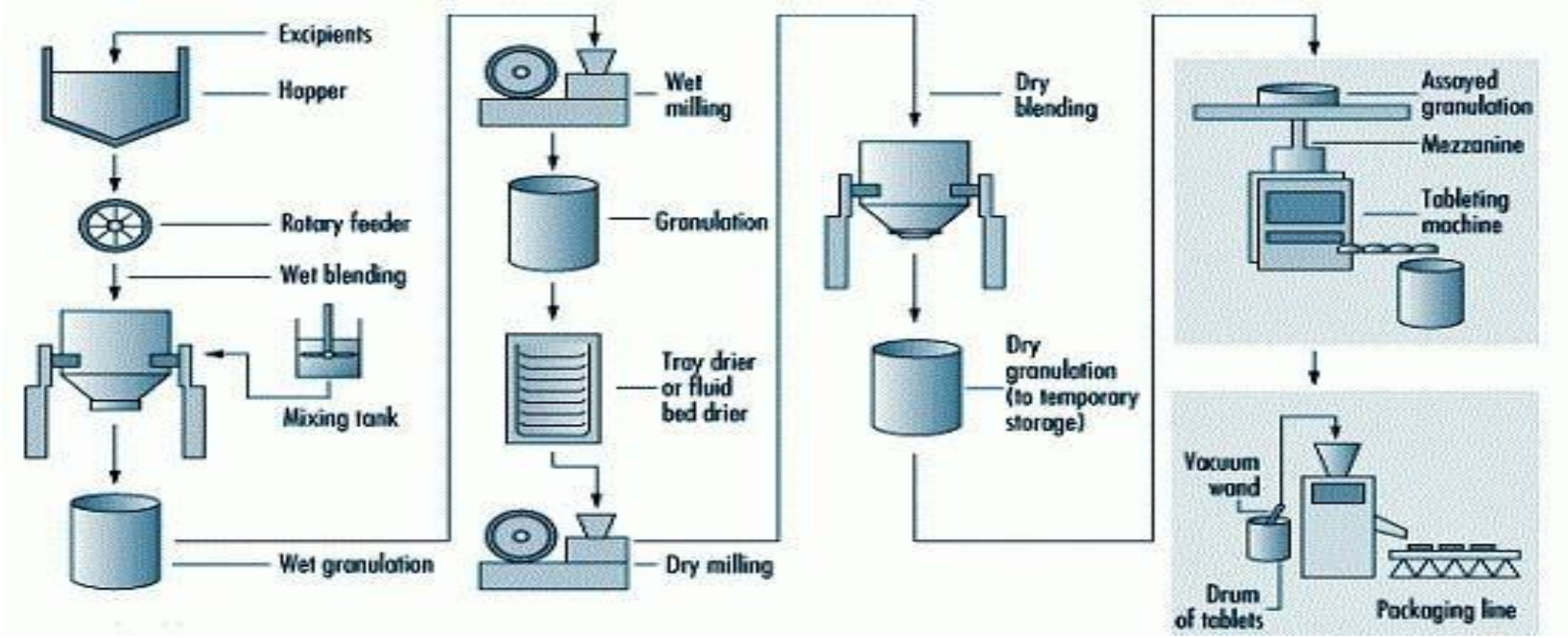
- Limited to drugs with **good flow & compressibility**
- Risk of **content uniformity problems**
- Segregation may occur

Application of Direct Compression Methods

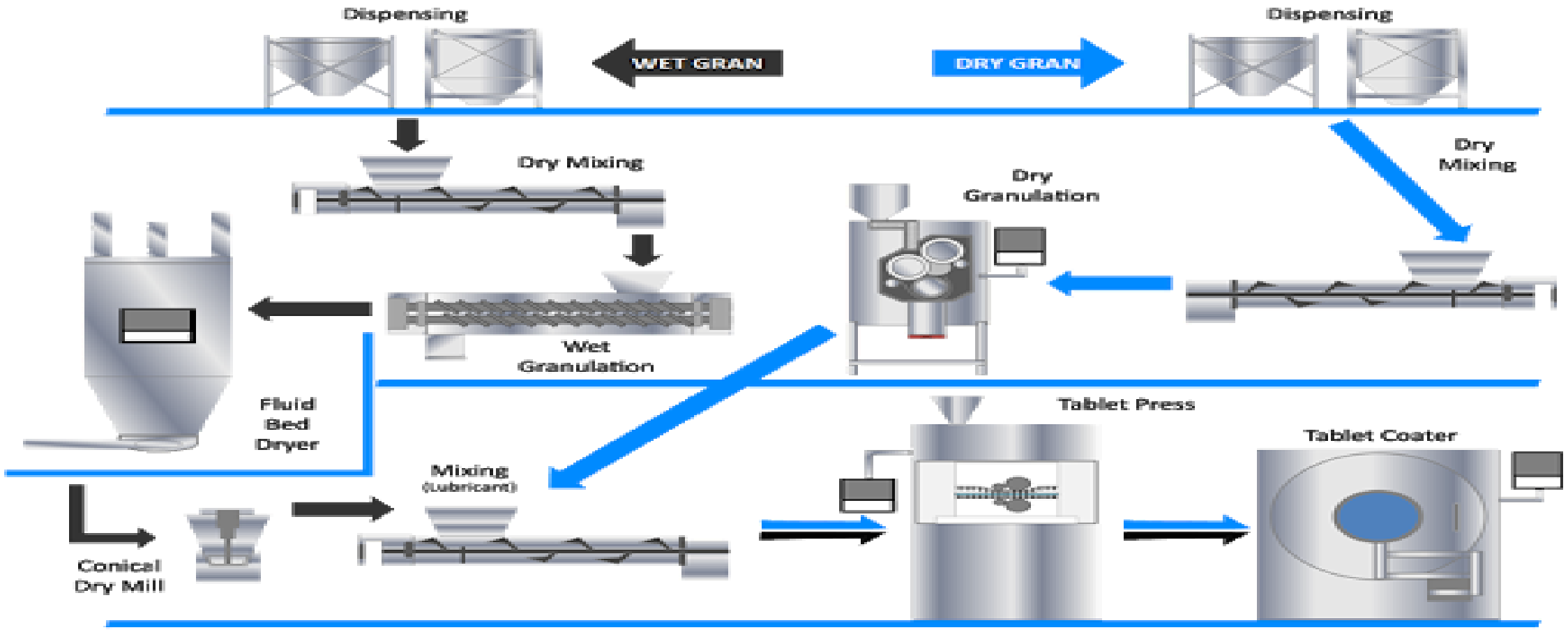
Applications

- Paracetamol tablets
- Antacid tablets
- Nutraceutical tablets

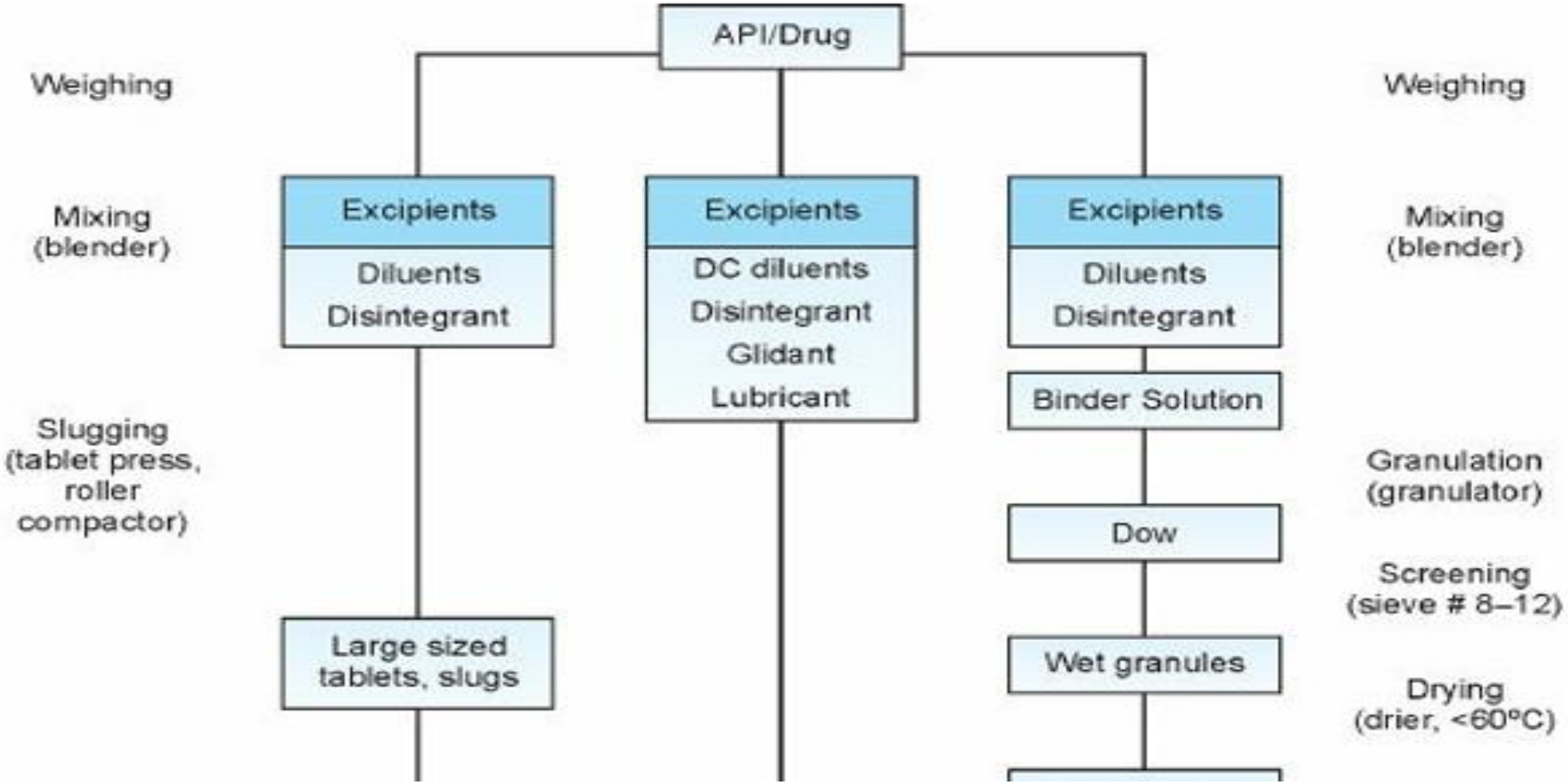
Overview of tablet manufacturing process



Overview of tablet manufacturing process



Granulation Methods



Granulation Methods

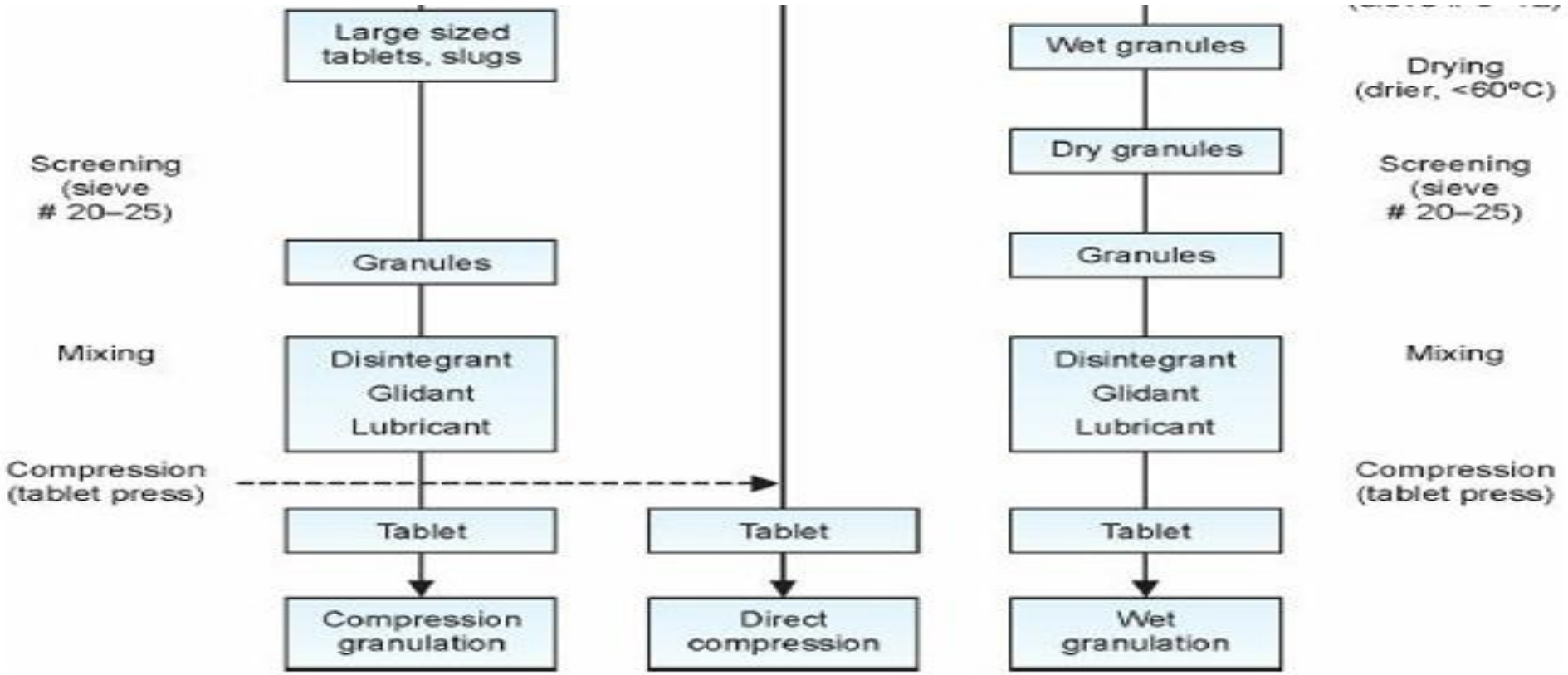


Fig. 13.2: Processing steps commonly required in the various tablet preparation techniques

Granulation Methods

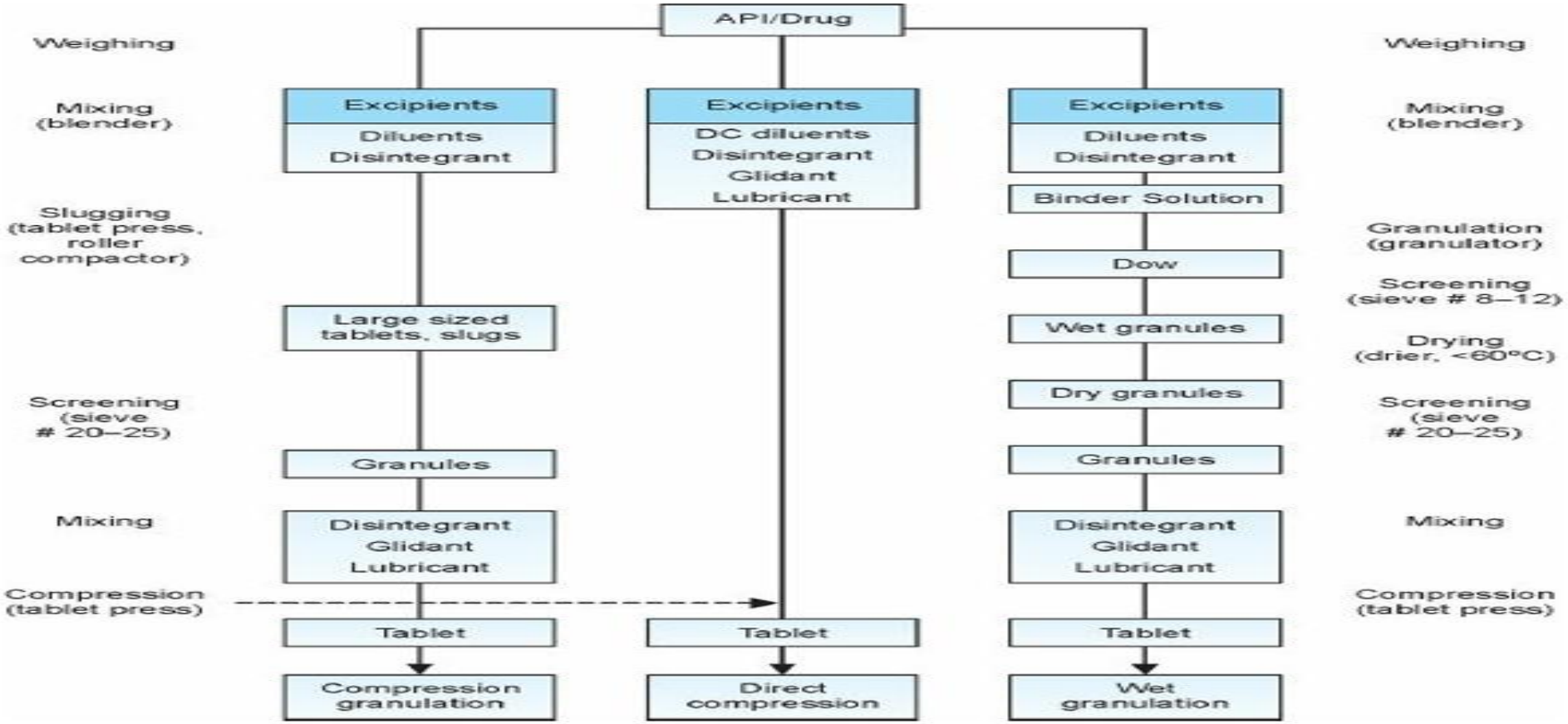


Fig. 13.2: Processing steps commonly required in the various tablet preparation techniques

Granulation Methods

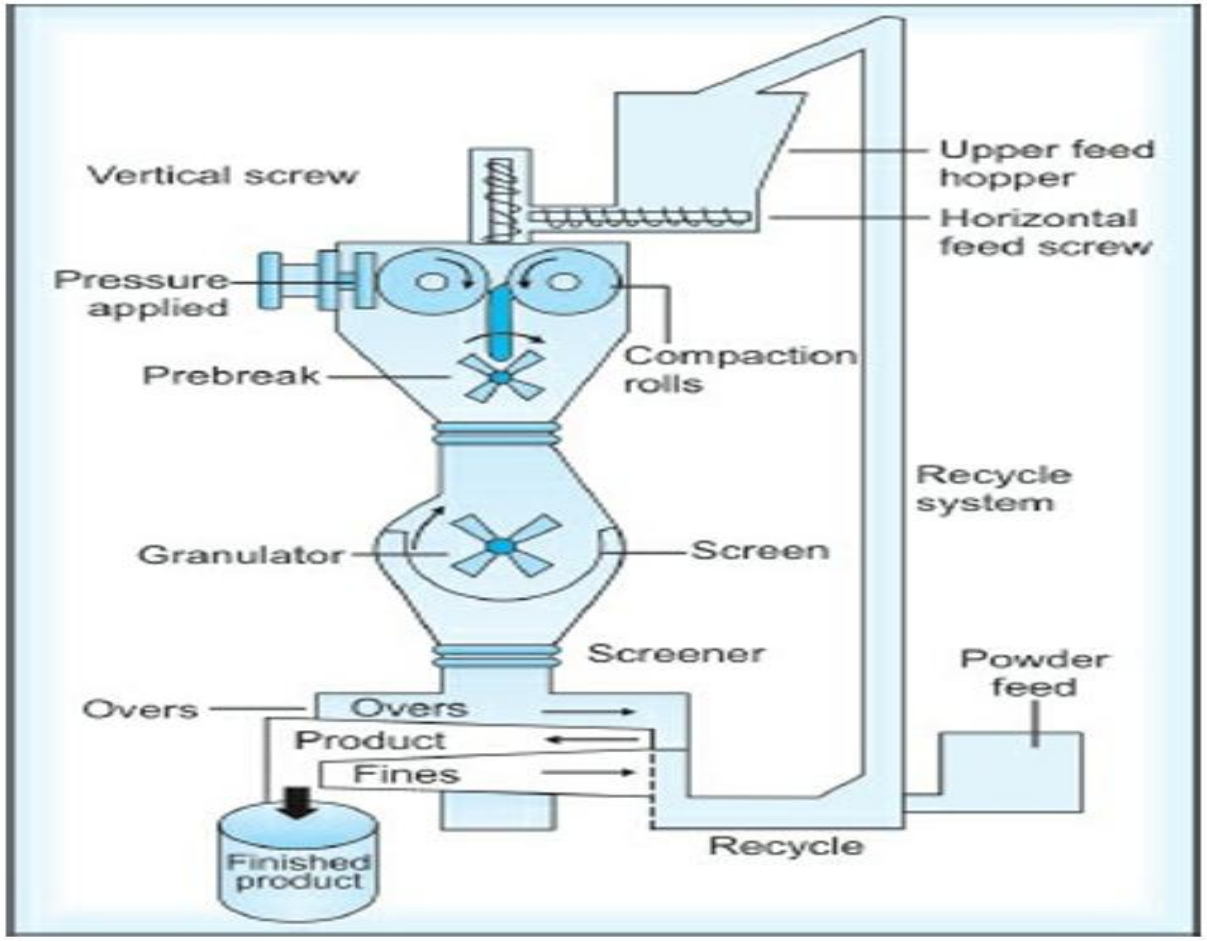
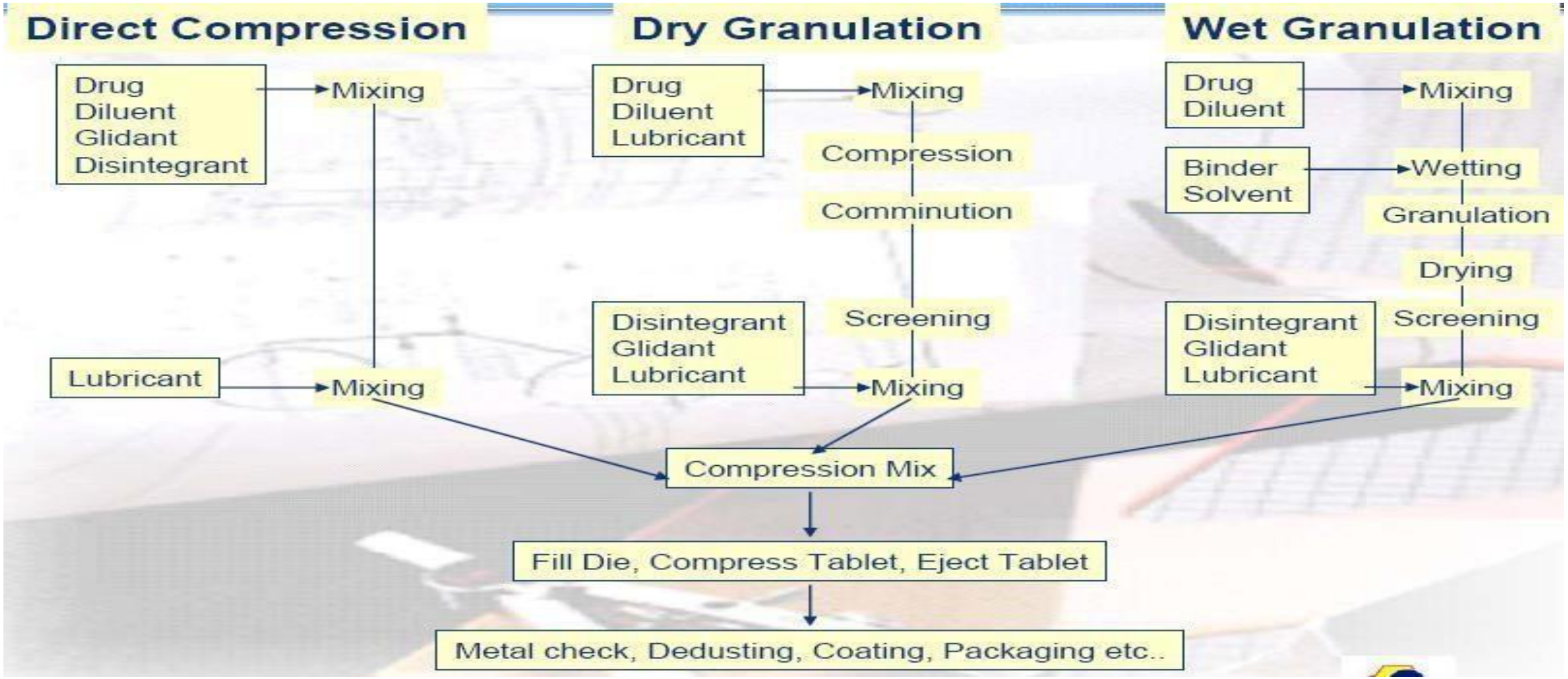


Fig. 13.3: Schematic diagram of a Chilsonator roller compactor in a granulation production system

Granulation Methods

The 16-Station rotary tablet press (*Courtesy* of ERWEKA)

Granulation technology on large scale



Steps in granulation method of tablet production

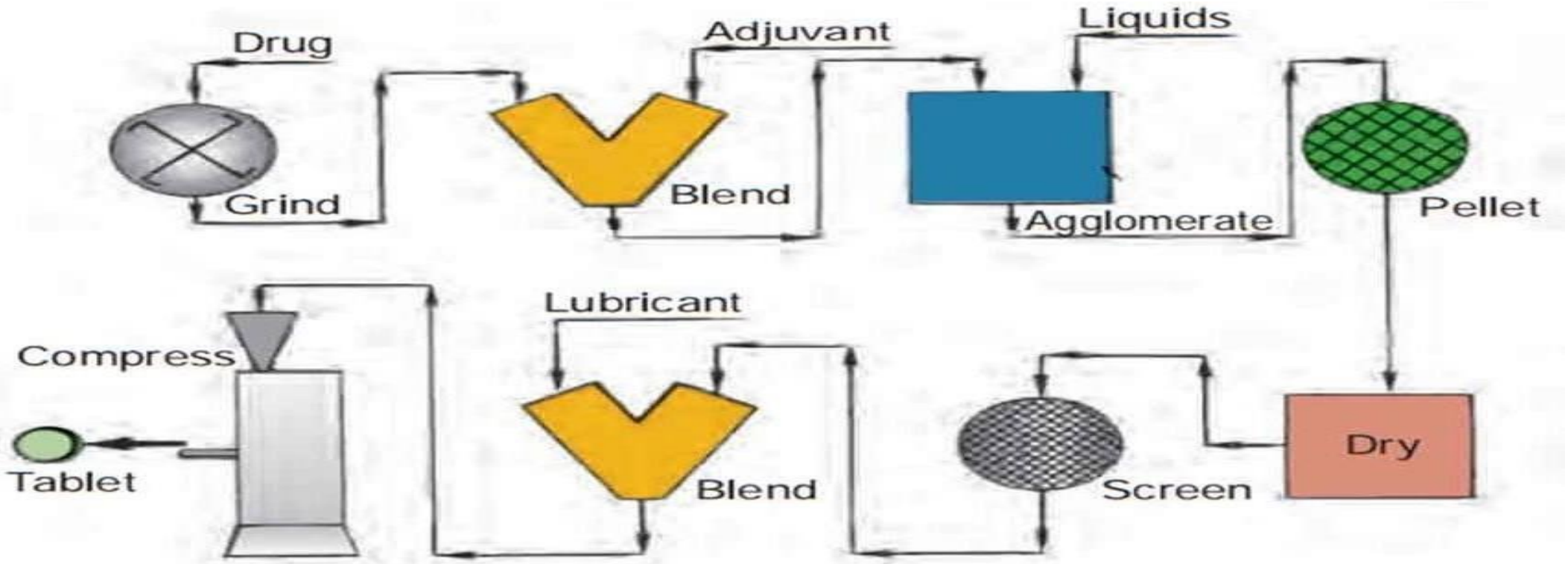


Photo credit: Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems. Philadelphia: Lipincott Williams and Wilkins

Steps to make powder ready for compression

- Weighing
- Mixing
- Granulation
- Screening
- Drying
- Screening
- Lubrication
- Compression

Compression of Tablets

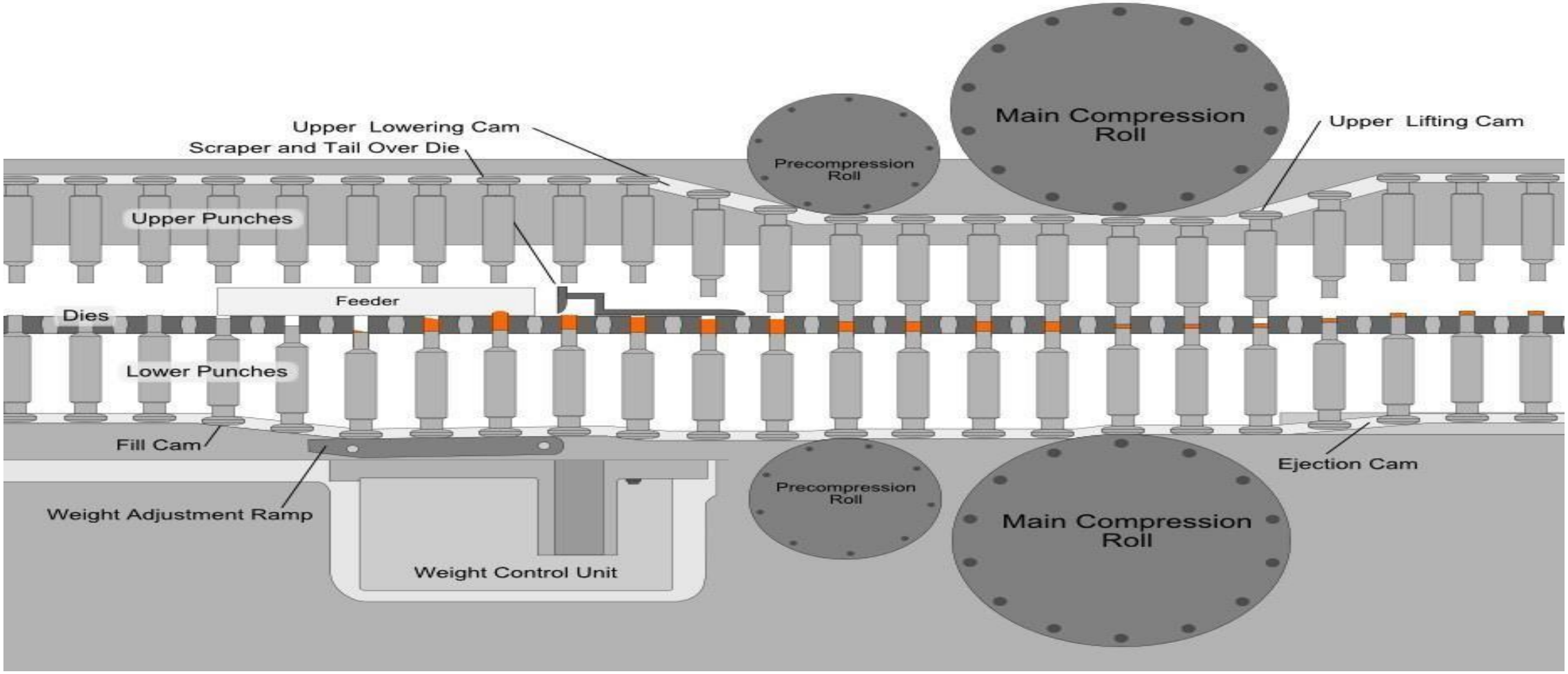
Tablet Compression

Performed using single punch or rotary tablet presses. Tablets are formed by applying high pressure to fill material in a die cavity.

Stages:

1. **Filling of die cavity**
2. **Compression of material between punches**
3. **Ejection of tablet**

Tablet compression stations



Compression Parameters

- **Hardness**: Mechanical strength
- **Weight variation**: Dose uniformity
- **Friability**: Resistance to abrasion

Marwadi University

THANK YOU!

MARWADI UNIVERSITY
Rajkot-Morbi Highway, Rajkot,
Gujarat, 360003, India.
www.marwadiuniversity.ac.in